

جامعة أم القرى
UMM AL-QURA UNIVERSITY

Course: Software Engineering Fundamentals

Project Title: Simple Budget Planner

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Section:14

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part1

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1. **Problem Context**

Many young people struggle with their financial situation because they weren't thought financial literacy nor how to properly plan their financial future. So , we decided to develop a program to help young people to start budgeting so they can pay their bills , and learn how to financially plan for their future.
This problem happen mostly with younger generations such Gen z because most of them are still university students or employees who have just started their jobs and don't know how to handle money yet.
Learning how to budget is important to avoid getting into debt , making irresponsible financial decisions and most importantly to not pass these bad spending habits to their kids.

2. **Target Users**

Most users of our program will be university students and young employees who need help managing their budget.

3- **Problem Statement**

A lot of university students and students employees do not find it easy to manage their income and expenses using traditional or unorganised ways. When they face this issue, they tend to spend more, they cannot save money, they make poor financial decisions etc. Which affects their finances. Furthermore, without constantly tracking expenses, identifying financial errors or managing the budget becomes more challenging to do so.

4- **Suggested Software Solution**

The proposed solution is a simple and easy-to-use budgeting system called Simple Budget Planner. It allows the user to record their earnings, track his daily and monthly expenses and watch the balance clearly. Similarly, it will notify the user when they are nearing their budget cap and warn about excessive spending in a particular category for users to make quicker and informed decisions regarding their finances.
The system helps improve financial awareness, develop saving behaviour, and monitor money with more order and confidence by presenting organized financial information and easy-to-use features.

5. **Project Scope**

Included Features
• Recording different sources of income (salary, allowance, part-time income).
• Adding and tracking daily and monthly expenses.
• Categorizing expenses (e.g., food, transportation, entertainment, bills).
• Automatically calculating and displaying the remaining balance.
• Setting a total monthly budget and/or individual budgets for each category.
• Sending alerts when the user is close to exceeding the planned budget.
• Providing a monthly summary report showing total income and total expenses.
• Visual charts or simple graphs to show spending patterns.
• Ability to edit or delete previously entered transactions.
• Secure user login system (basic authentication).

Excluded Features
• Direct integration with bank accounts or credit cards.
• Online payment or money transfer services.
• Investment or stock trading features.
• Advanced financial planning or professional financial advisory services.
• Complex loan or debt management systems.
• Multi-user business accounting features.
• Artificial intelligence–based financial predictions.

6. **Assumptions and Constraints**

Assumptions
• Users have access to a smartphone or computer to use the system.
• Users will enter their income and expense data regularly and accurately.
• Users have basic knowledge of using digital applications.
• The system is designed for personal use, not for businesses.
• Users are responsible for maintaining the privacy of their login information.

Constraints
• The project is limited by the time available during the academic semester.
• Development will be based on basic to intermediate programming skills.
• Advanced security systems may not be fully implemented due to limited resources.
• The first version may not support all devices or operating systems.
• Limited testing time may restrict full performance optimization.
• The system will operate without real-time bank data integration.

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7.The following timeline presents the main stages of developing the Simple Budget Planner system:

Week	Work Complete
week1	Defining the project problem and identifying target users
week2	Writing problem statement, solution, and project scope (Phase 1 submission)
week3	Identifying system requirements and creating user personas
week4	Creating use case diagram and submitting Phase 2
week5	Designing system architecture and interface wireframes
week6	Creating UML diagrams and finalizing system design
week7	Reviewing design and submitting Phase 3
week8	Developing prototype, testing system, and submitting Phase 4

Student	Task Completed
Layan Saleh Al Qarni	1.Problem Context 2.Target Users
Ghadi Bandar Mukhtar	3.Problem Statement 4.Proposed Software Solution
Yara Shadi Al Ansari	5.Project Scope 6.Assumptions and Constraints
Ghala Hussein Al Sharif	7.Project Development Timeline
Joud Abdullah Al osaime	Prepared the project report.
anhar mohammed al nafiei	Reviewed and edited the final project report